

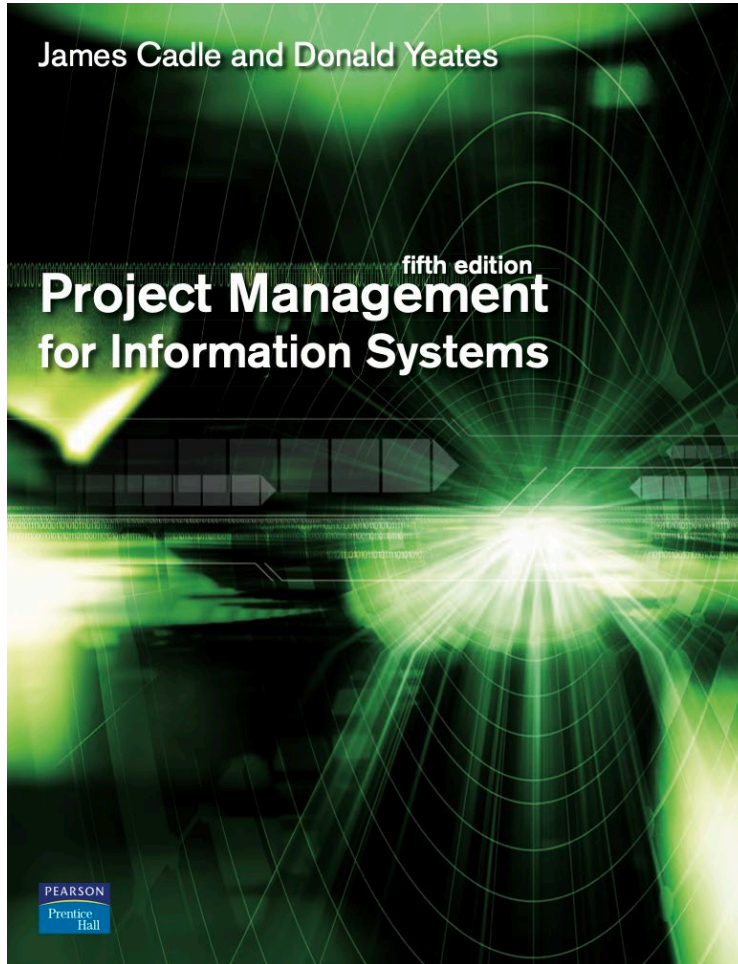
BU5300 – IT Project Management

Lecture 4: Agile project management

Based on notes by Prof. Knut H. Rolland

Overview of the Course

Lectures and Book Chapters



Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - *Agile Development*

Lecture 5 - Chapters 7 and 8

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11, 12 and 13

Lecture 8 - Chapters 14 and 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapters 20, 22 and 23

IMPORTANT! Changes may occur. Always check on Canvas and TimeEdit

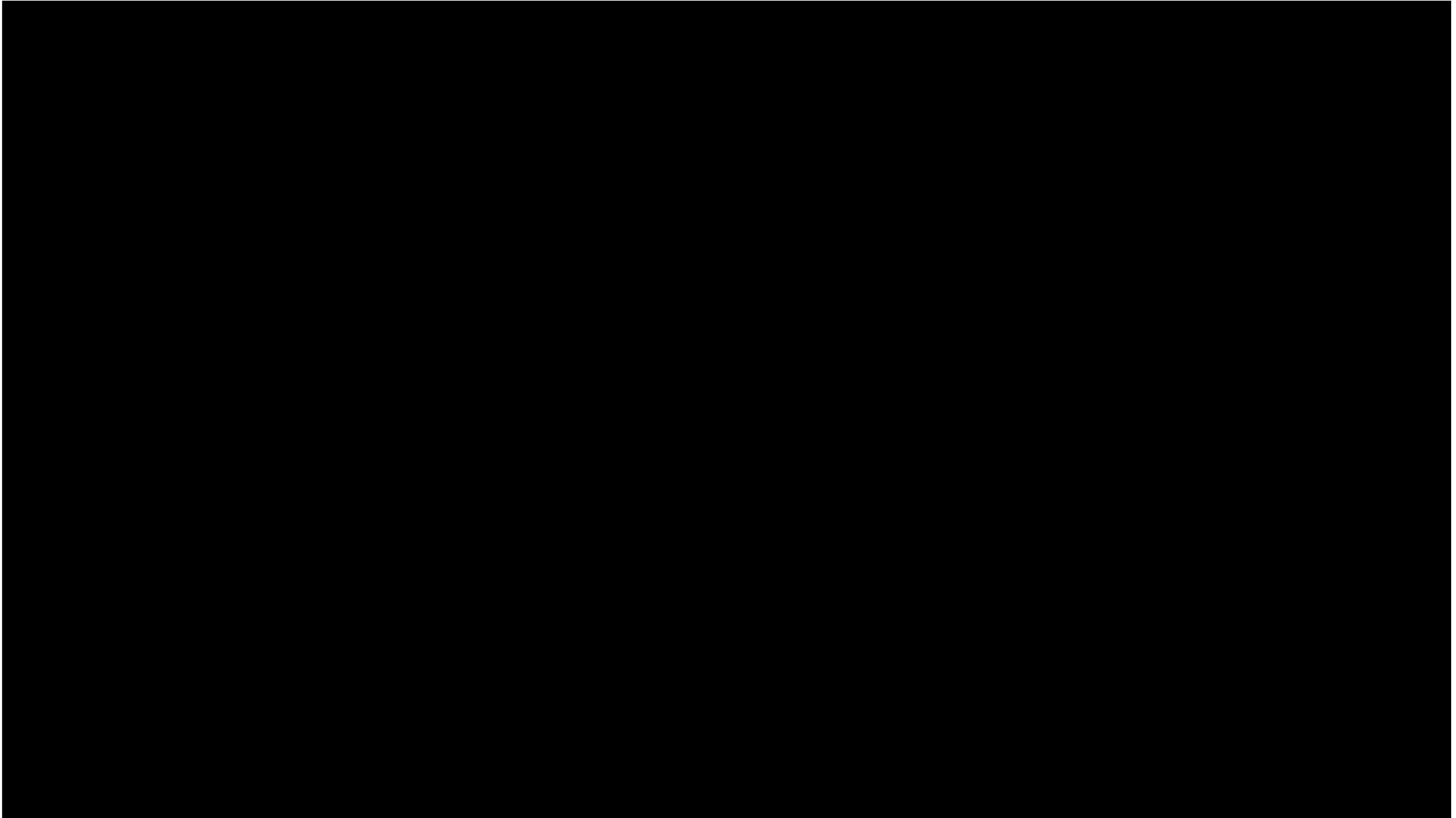
Agile software development

Manifesto for agile software development (2001) <http://agilemanifesto.org/>

Agile values...

- **Individuals and interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

Agile Video from O'Reilly Learning



Some core characteristics

- Led by experienced practitioners, not researchers or consultants
- Emphasizes minimal, just-in-time planning
- Encourages continuous learning and improvement
- Utilizes small, co-located teams
- Balances physical artifacts and software tools for coordination
- Prioritizes close customer interaction
- Supports self-managed teams with decentralized decision-making
- Aligns with workplace democracy and flatter hierarchies (decentralized decision)
- Promotes generalization of work tasks for all team members
- Allows flexibility in changing requirements

Research on IT projects in Norway

Table 2. The relationship among budget size category, development method, and project performance.*

Project performance	Development method	Small (n = 120)	Medium (n = 54)	Large (n = 22)
Successful (n = 31)	Agile	19%	24%	7%
	Nonagile	0%	19%	0%
Acceptable (n = 102)	Agile	65%	58%	50%
	Nonagile	19%	31%	25%
Failed (n = 13)	Agile	2%	3%	14%
	Nonagile	23%	6%	13%

**The percentages are the proportion of successful, acceptable, and failed projects for projects of the same budget size, category, and development method.*

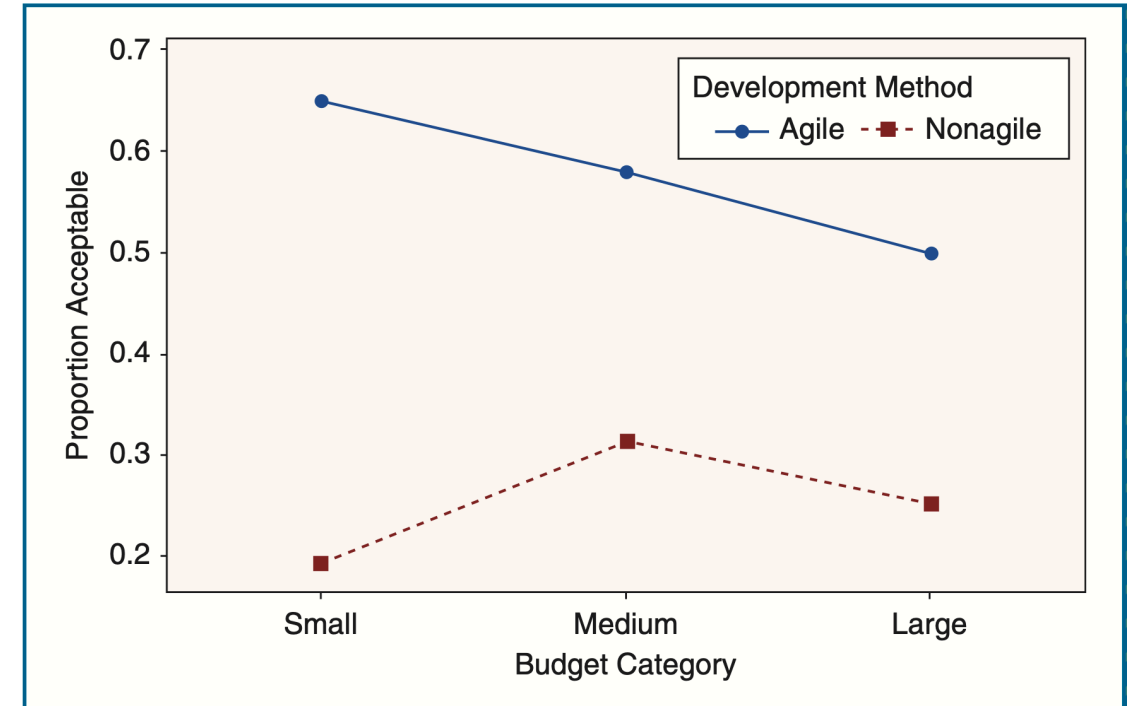
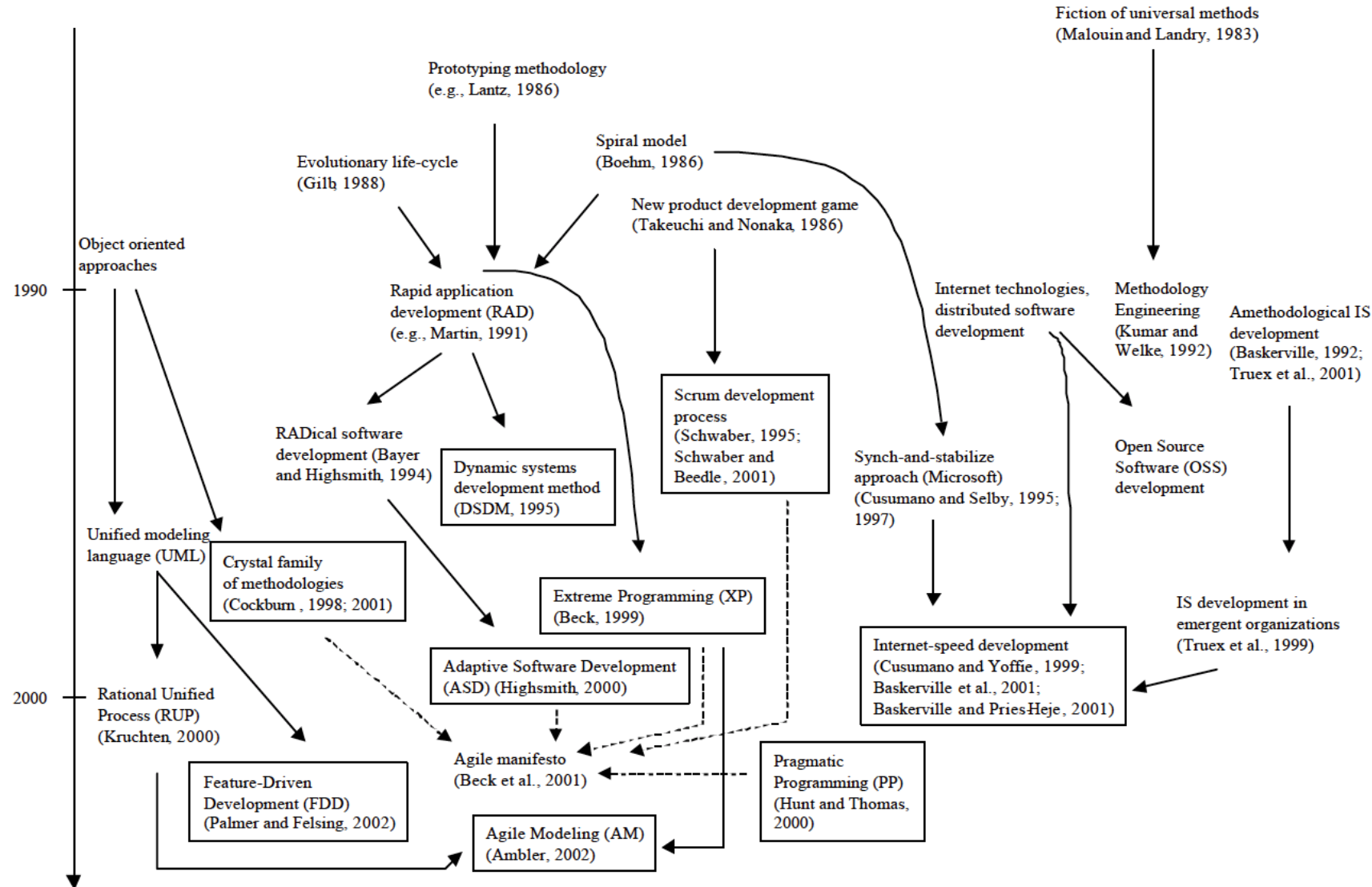


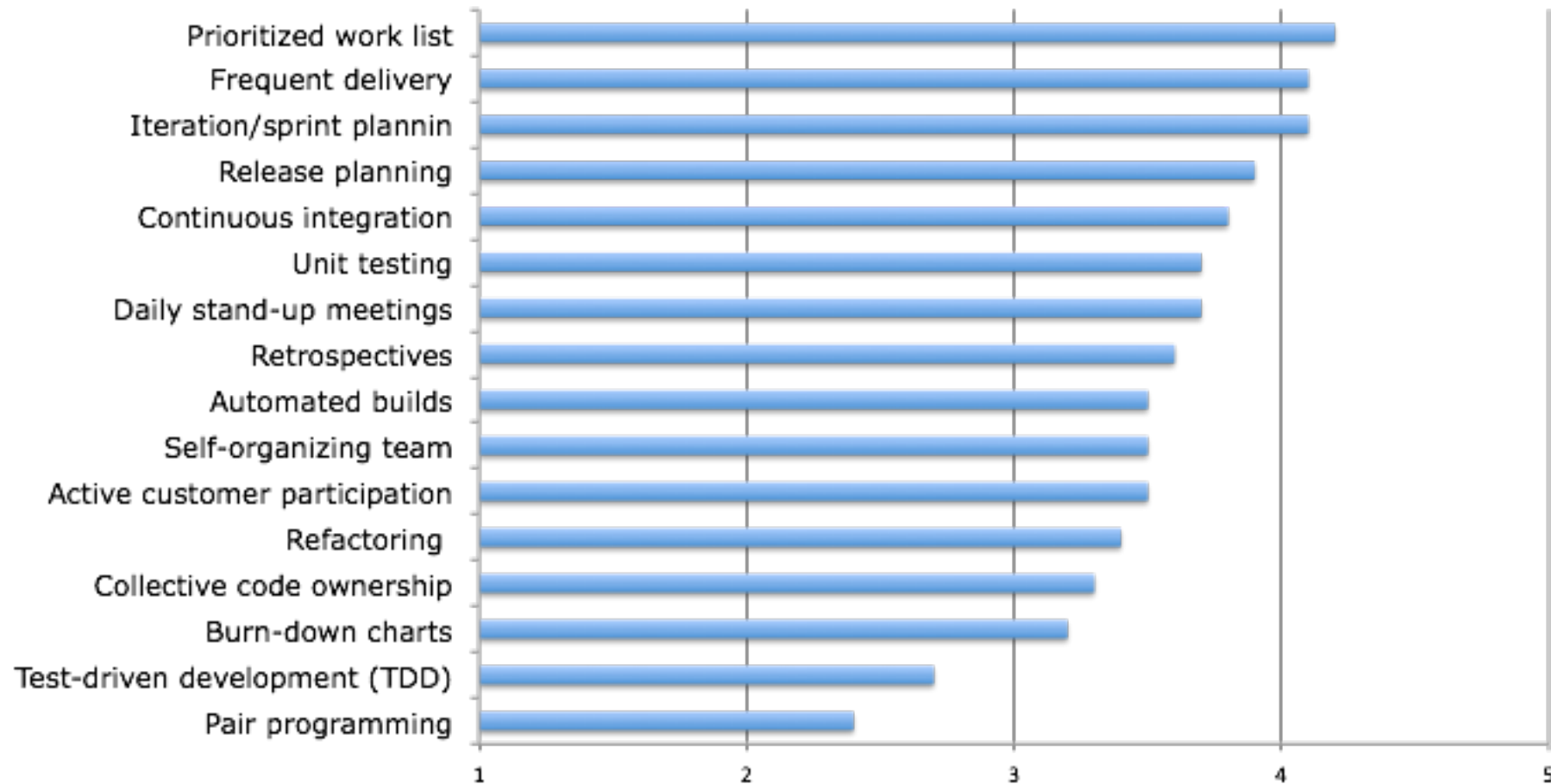
FIGURE 1. The interaction plot of projects with acceptable performance.

Jørgensen, Magne. **Relationships between project size, agile practices, and successful software development: results and analysis.** IEEE Software 36.2 (2019): 39-43.

New Directions on Agile Methods



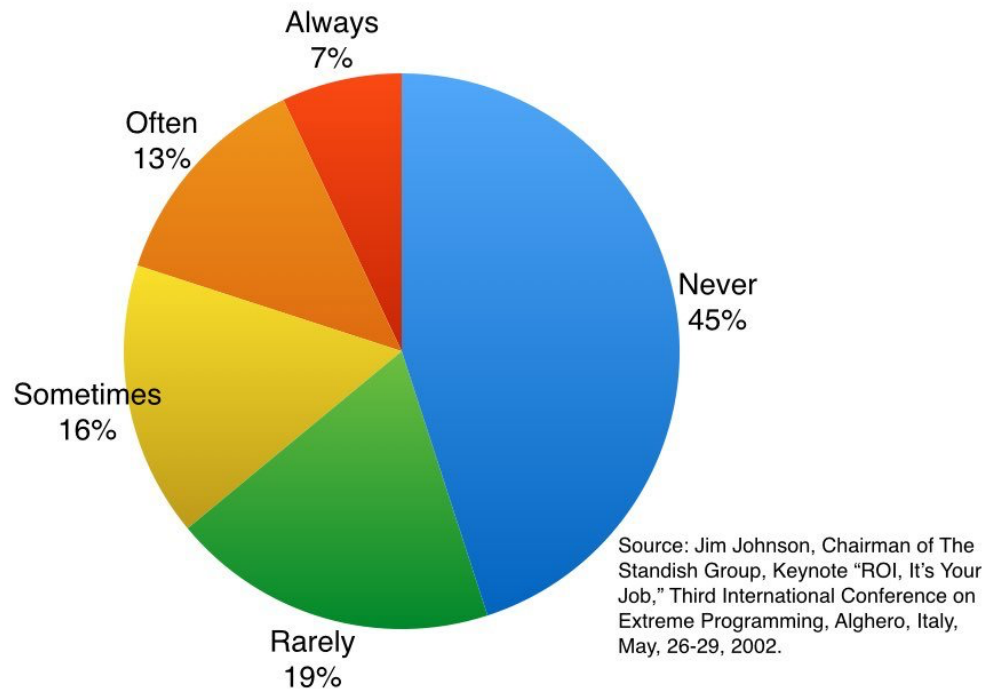
Use of agile practices



Rodríguez, Pilar, et al. **Survey on agile and lean usage in finnish software industry**. Proceedings of the 2012 ACM-IEEE International Symposium on Empirical Software Engineering and Measurement. IEEE, 2012.

Does not pay off to develop all features

Feature Use in Four Internal-Use Products



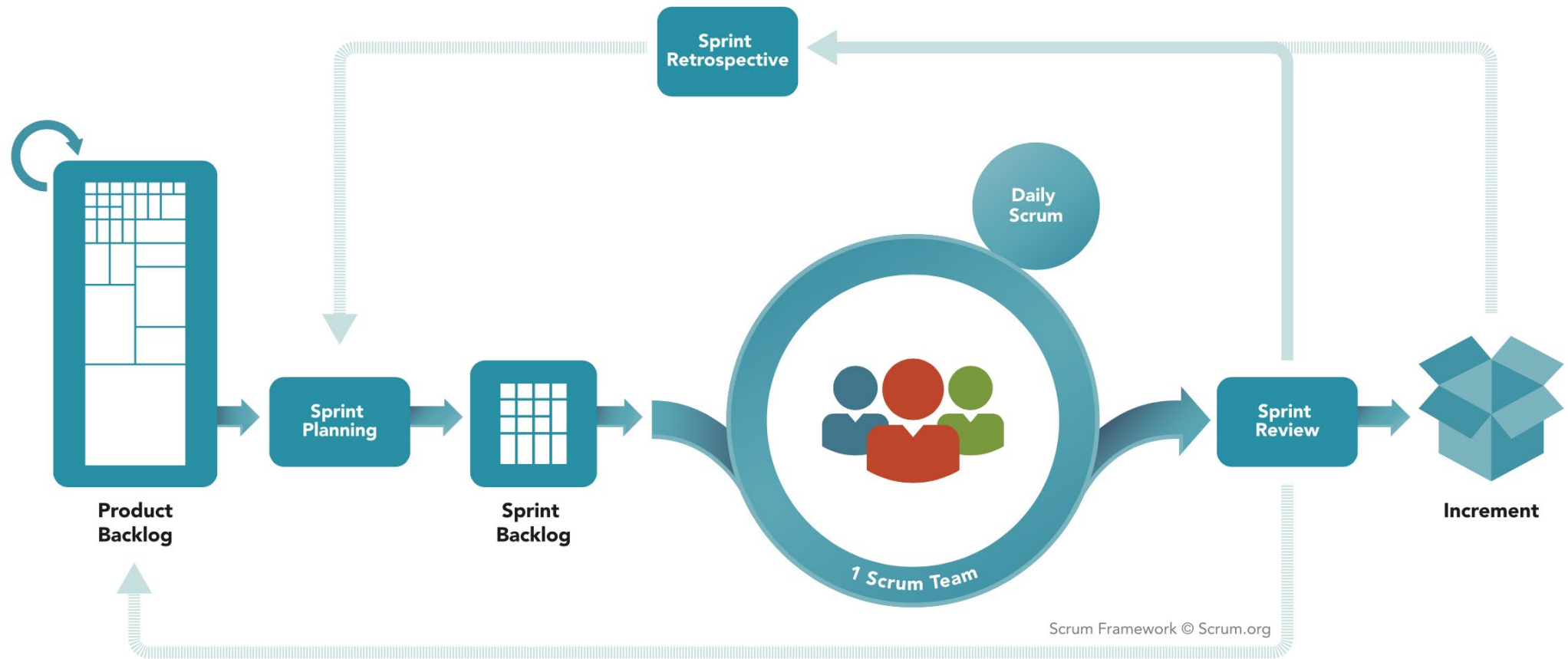
- If you want to go faster, stop doing the 45% of features that are never used
- Also, the Pareto rule: 20% of the system provides ~80% of the value

Scrum (Agile Framework)

Scrum

- Created by Jeff Sutherland and Ken Schwaber in the early 1990s
- Rooted in empirical process control and complex adaptive systems
- Derived from rugby football terminology
- Promotes self-organizing teams for problem-solving
- Comprises roles, ceremonies, and artifacts
- Widely adopted in software development
- Evolves continuously during projects

Scrum

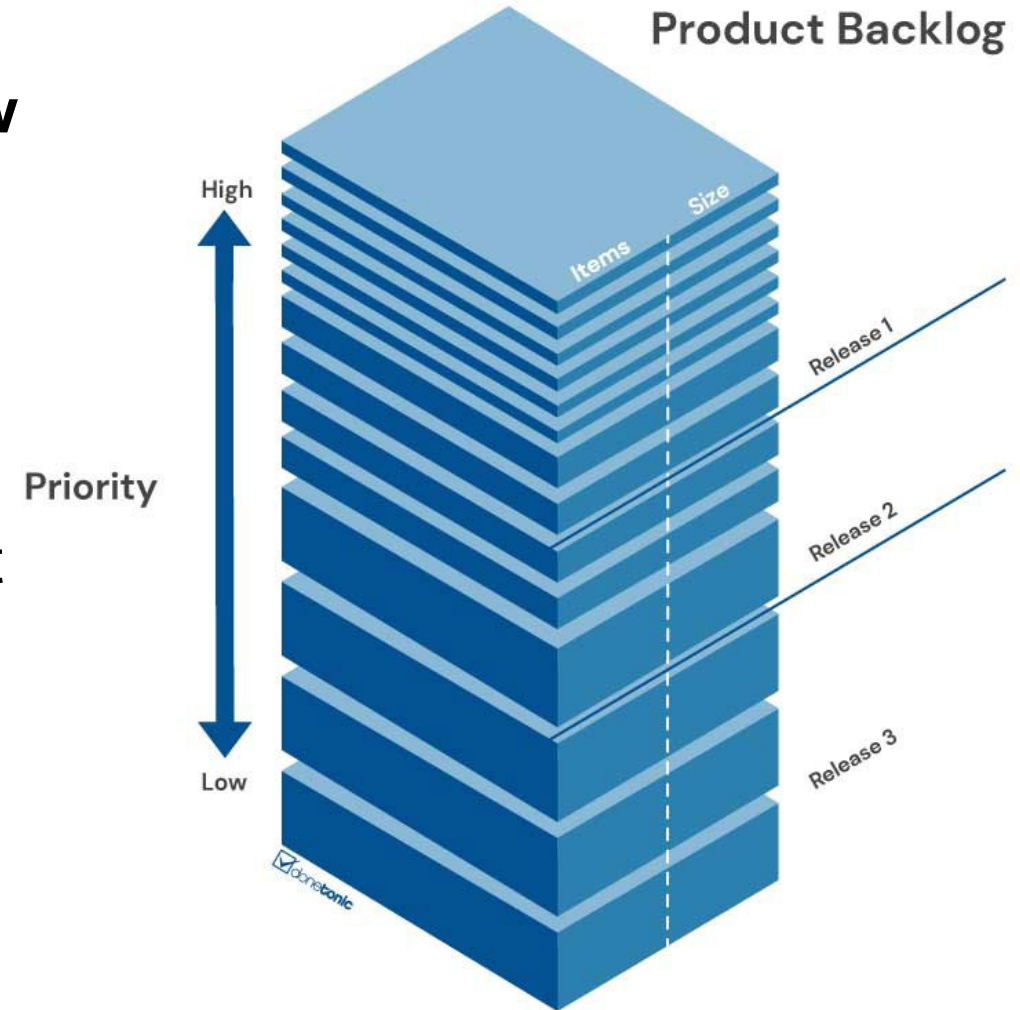


Source: www.scrum.org

Scrum

Product Backlog

- An overview of all items (PBIs) that exist **now**
 - Functional requirements
 - Research items (technology, market, etc)
 - Changes
 - Bugs
- **Product owner** can add new items to the list
- **Sprint backlog** is a subset of PBIs for the current sprint
 - Committed to deliver
 - Make up the sprint planning



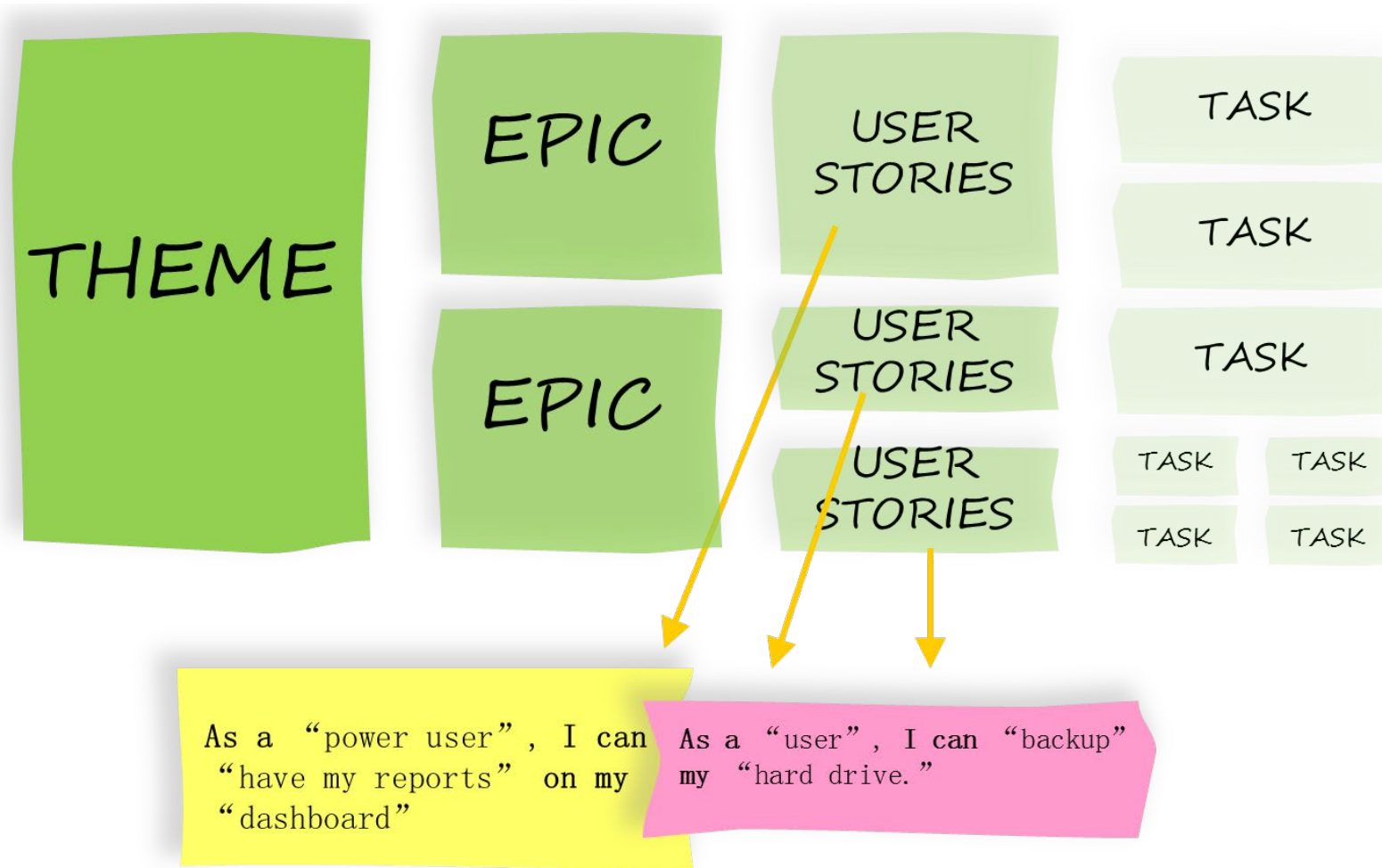
Scrum

Sprint planning

1. Define a goal for the sprint
2. Take the product backlog as a starting point
3. Take into account what the product owner thinks
4. Break things from the product backlog down into the sprint backlog
5. Feel free to describe functional requirements with user stories
6. Decide how to reach the goal

User story

A way to describe user needs



User story

A way to describe user needs



As a <role>
I want <goal>
so that <benefit>

Acceptance criteria:

(Conditions of Satisfaction)

...

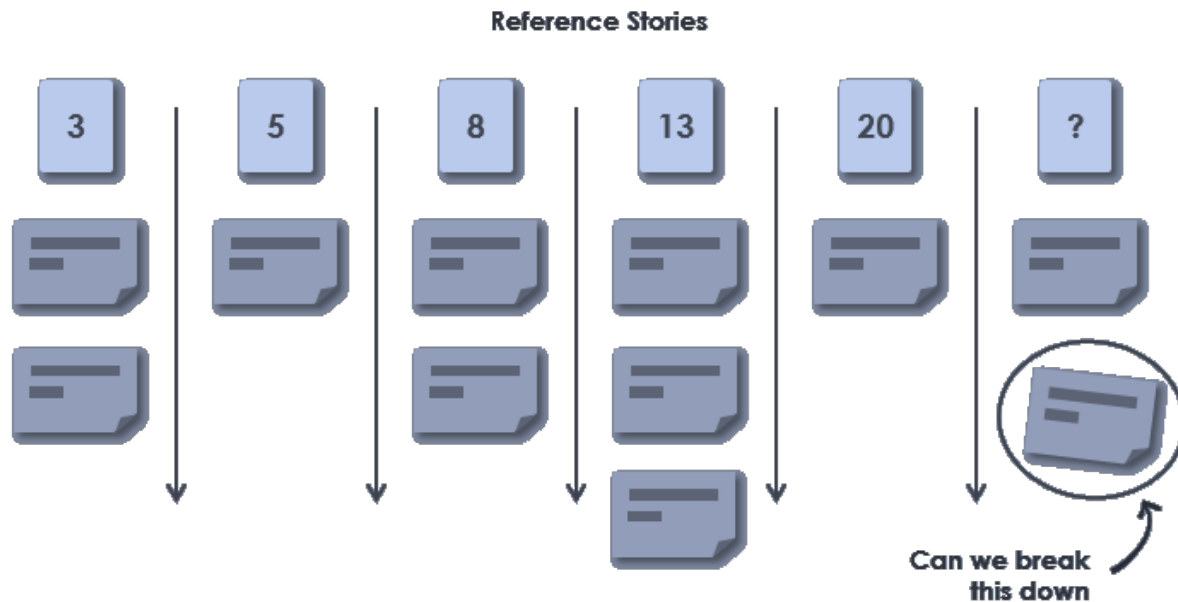
...

- As a **user**
- I want **register my account**
- so that **I can access VisitOslo STN**

- **Acceptance criteria:**
 - A user fill up the registration form
 - The phone number is validated
 - The user is notified that its registration is accepted

Story Points

How do we know how much effort a PBI requires?



- Story points are estimates of effort
 - Added to user stories when detailing them
 - Work-hours are absolute, SP are relative
- The user stories must be small
 - Less than features, much less than epics
 - If your sprint has 20SPs, that is the maximum a task can be
 - Larger stories should be broken down into smaller PBIs

Scrum

Daily meeting

- Every day!
- Maximum 15 minutes
- Everyone stands
- Answer three things:
 - What did you do yesterday?
 - What are you going to do now?
 - Is there any impediment, i.e., something stopping you from moving forward?



Scrum

Sprint review and retrospective

Sprint Review

- Focus on the product
- Review the delivery (what is done)
- Update the product backlog
- Prepare for next sprint planning
- Meeting at the last day of the sprint

Sprint Retrospective

- Focus on the process
- Identify what went right
- And what can be improved (and how)
- Improve culture, boost the team
- After the sprint review

Agile Planning

Visit Oslo's STN example

PBI	Sprint #	Sprint Planning	Sprint Backlog	Sprint Review
Project setup & ways of working	1	Establish repo, CI, branching, DoD, team rituals, basic architecture	Create repo + CI; lint/format; env setup; backlog refinement; Definition of Done	Done (Sprint 1)
Basic UI shell + navigation	1	Deliver a working skeleton app with core pages/routes	App layout; routing; header/footer; placeholder pages; basic styling	Done (Sprint 1)
Data model + content structure (attractions/events)	2	Agree on domain model + sample dataset	Define schemas; seed data; JSON/API contract draft; mapping to UI components	In progress
List view for attractions/events	2	Users can browse a list with basic filtering	List component; filter controls (basic); empty/error states	In progress
Detail page for attractions/events	3	Not started	...	Not started

Project Manager in Agile

Project manager's role

- Reach your project goals in terms of budget, quality and time
- Ensure that the project has the right expertise
- Establish teams and motivate good efforts
- Manage conflicts and communication within the team and between teams around the outside world
- Plan and track your project
- Make decisions



Agile project management approaches

- Co-located and small teams
- Coordination through physical artifacts as well as software tools
- Close interaction with customer
- Self-managed teams

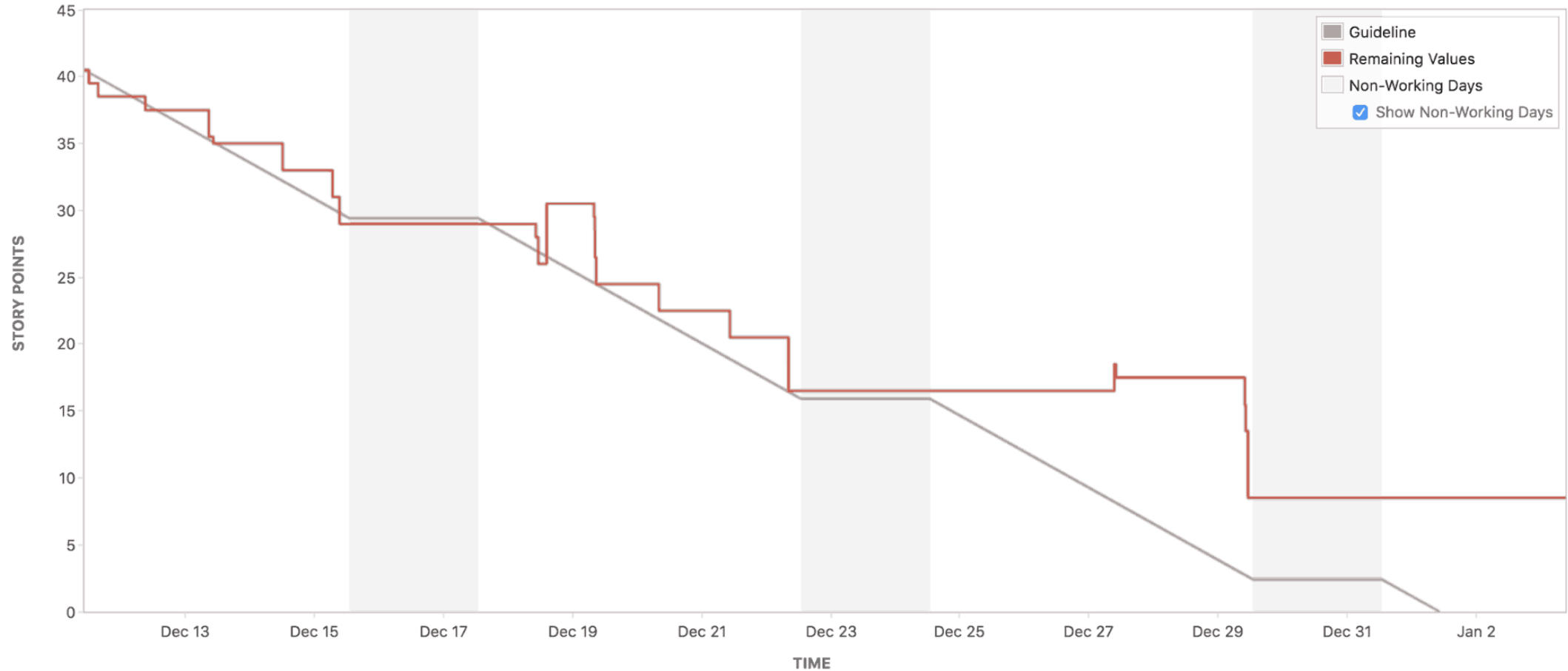
Final Definition of Agility

the continual readiness of an ISD method to rapidly or inherently create change, proactively or reactively embrace change, and learn from change while contributing to perceived customer value (economy, quality, and simplicity), through its collective components and relationships with its environment.

Conboy, Kieran. **Agility from first principles: Reconstructing the concept of agility in information systems development.** Information systems research 20.3 (2009): 329-354.

Burn-down charts

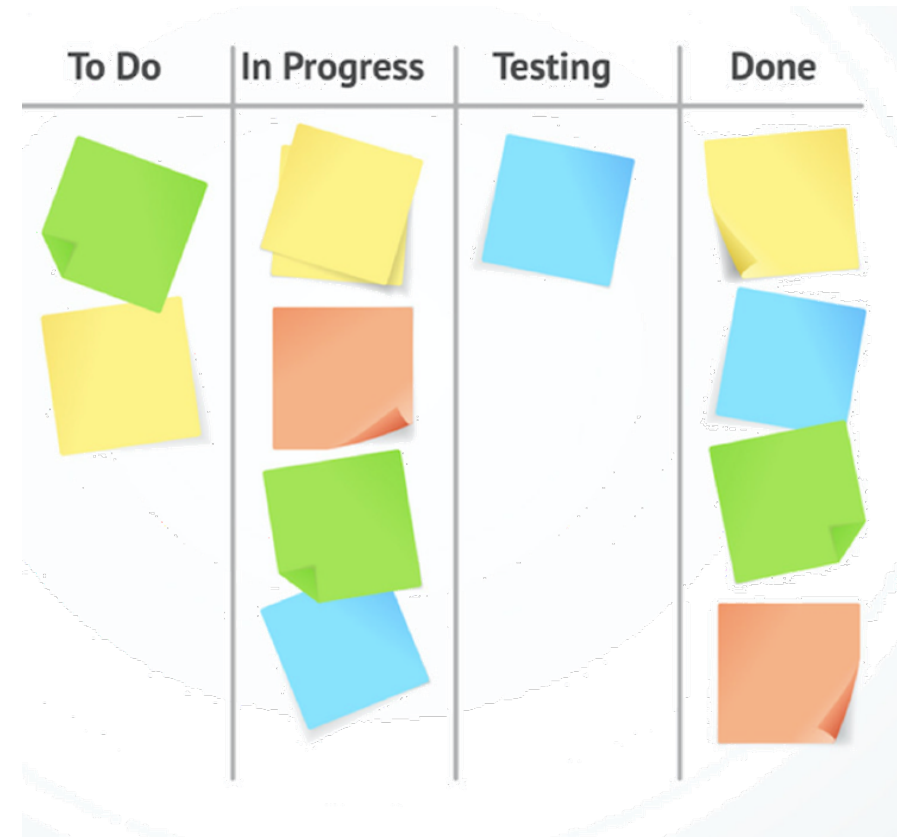
A tool for managing sprints



Kanban Board

A tool for managing agile processes

- It helps visualize work, progress, and assess efficiency
- Every work item is represented as a separate card on the board.
- As work is progressing, team members move the cards to the right columns



Digital tools for Agile

The image displays two digital tools used for Agile project management: Trello and Confluence.

Trello Interface (Left):

- Product Backlog (10 items):** Navigation Menu (5), Backgrounds (2), Scenes Landing Page (3), Scenes Navigation, Scene Detail.
- Sprint Planning (12 items):** Data storage (3), Character (5), Character (2), Character (2).
- Current Sprint:** Add a card...
- In Progress (3.56 items):** Basic Design (2, 1).
- Done (1 item):** Bootstrap framework (1).

Confluence Interface (Right):

- Board:** Quick Filters, Release, ...
- TO DO 5:**
 - Engage Jupiter Express for outer solar system travel (SPACE TRAVEL PARTNERS, 5, TIS-25)
 - Create 90 day plans for all departments in the Mars Office (Local Mars Office, 9, TIS-12)
 - Engage Saturn's Rings Resort as a preferred provider (Space Travel Partners, 3, TIS-17)
 - Enable Speedy SpaceCraft as the preferred
- IN PROGRESS 5:**
 - Requesting available flights is now taking > 5 seconds (SeeSpaceEZ Plus, 3, TIS-8)
 - Engage Saturn Shuttle Lines for group tours (Space Travel Partners, 4, TIS-15)
 - Establish a catering vendor to provide meal service (Local Mars Office, 4, TIS-15)
 - Engage Saturn Shuttle Lines for group tours
- CODE REVIEW 2:**
 - Register with the Mars Ministry of Revenue (Local Mars Office, 3, TIS-11)
 - Draft network plan for Mars Office (Local Mars Office, 3, TIS-15)
- DONE 8:**
 - Homepage footer uses an inline style - should use a class (Large Team Support, 1, TIS-68)
 - Engage JetShuttle SpaceWays for travel (Space Travel Partners, 5, TIS-23)
 - Engage Saturn Shuttle Lines for group tours (Space Travel Partners, 1, TIS-15)
 - Establish a catering vendor to provide meal service (Local Mars Office)

Trello

Confluence

Principles of agile project management

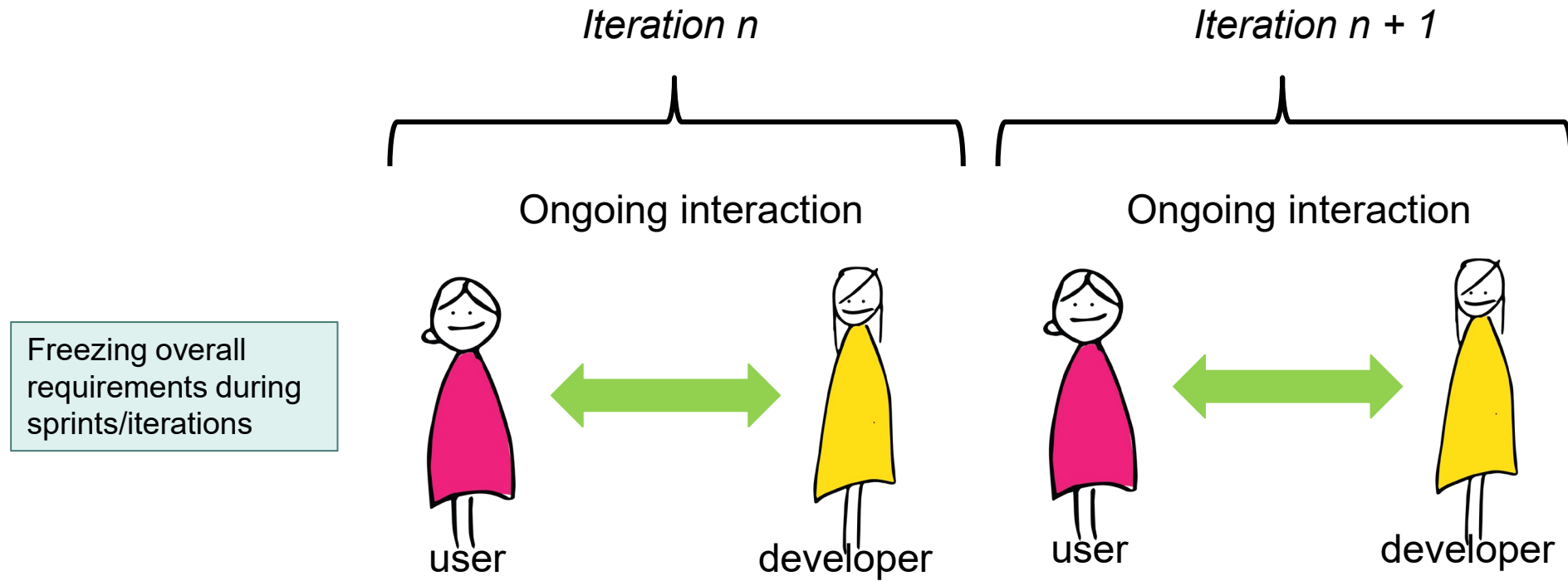
- **Minimum critical specification:** should not specify more than is absolutely necessary and critical to overall success
- **Autonomous teams:** autonomous teams are responsible for managing and monitoring their own processes and executing tasks
- **Redundancy:** Team members should be skilled in more than one function
- **Feedback and learning:** Feedback and learning are integral to project execution and the project's interaction with the environment

Lean and its principles

TPS – “Toyota Production System” principles that are translated to software development

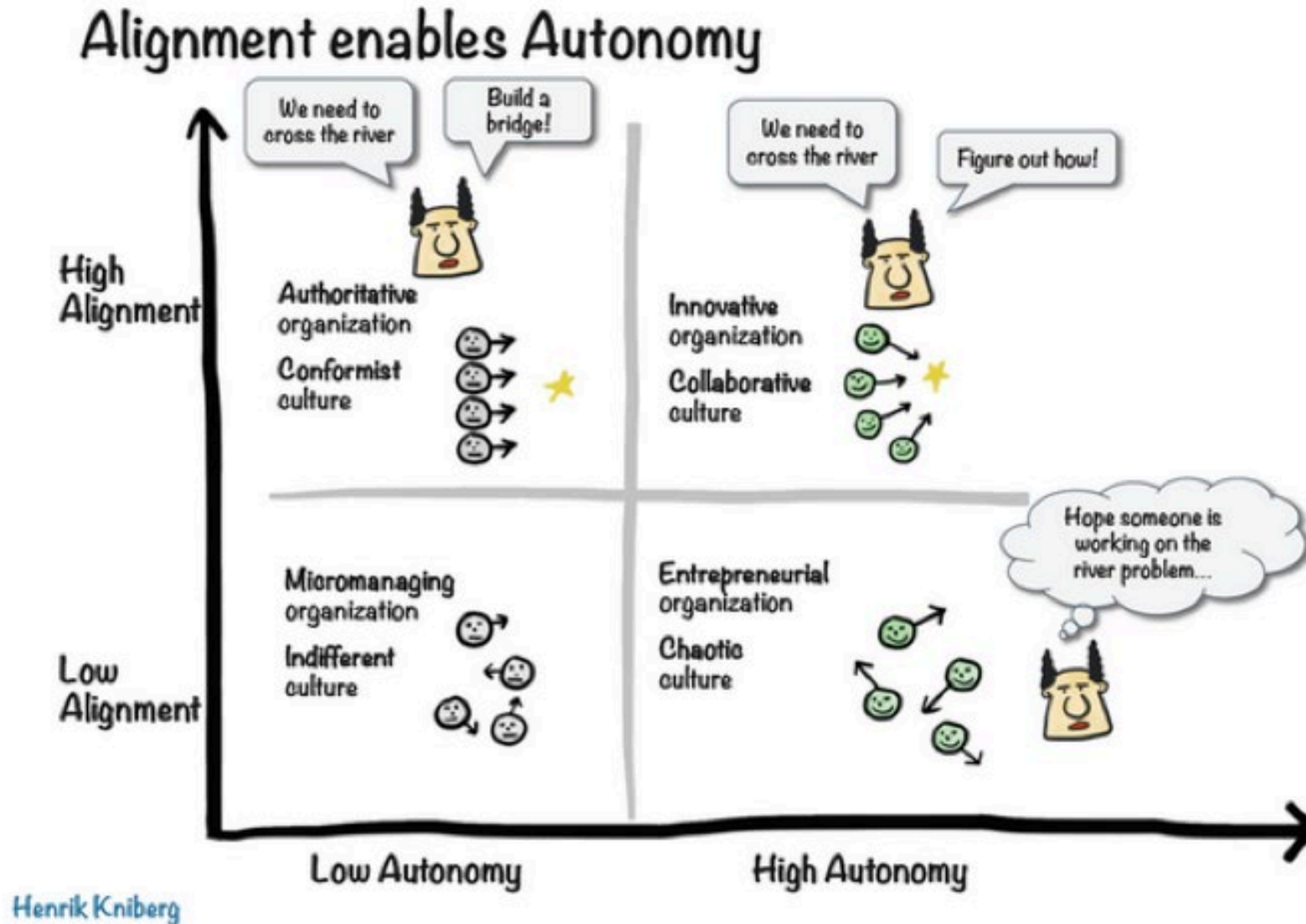
- **Optimizing the Whole:** Focus on the entire value stream, deliver a complete product
- **Eliminate Waste:** Waste is anything that does not add customer value
- **Build Quality In:** modularisation and test-first development
- **Learn constantly**
- **Deliver fast:** small batches and few things in process
- **Engage everyone:** semi-autonomous teams, end-to-end responsibility
- **Keep getting better:** Failure is a learning opportunity

Minimal specification – and as late as possible



Specifying as an ongoing process

Autonomous teams



Illustrated by [Henrik Kniberg](#) in the “[Spotify Engineering Culture](#)” and several of his talks.

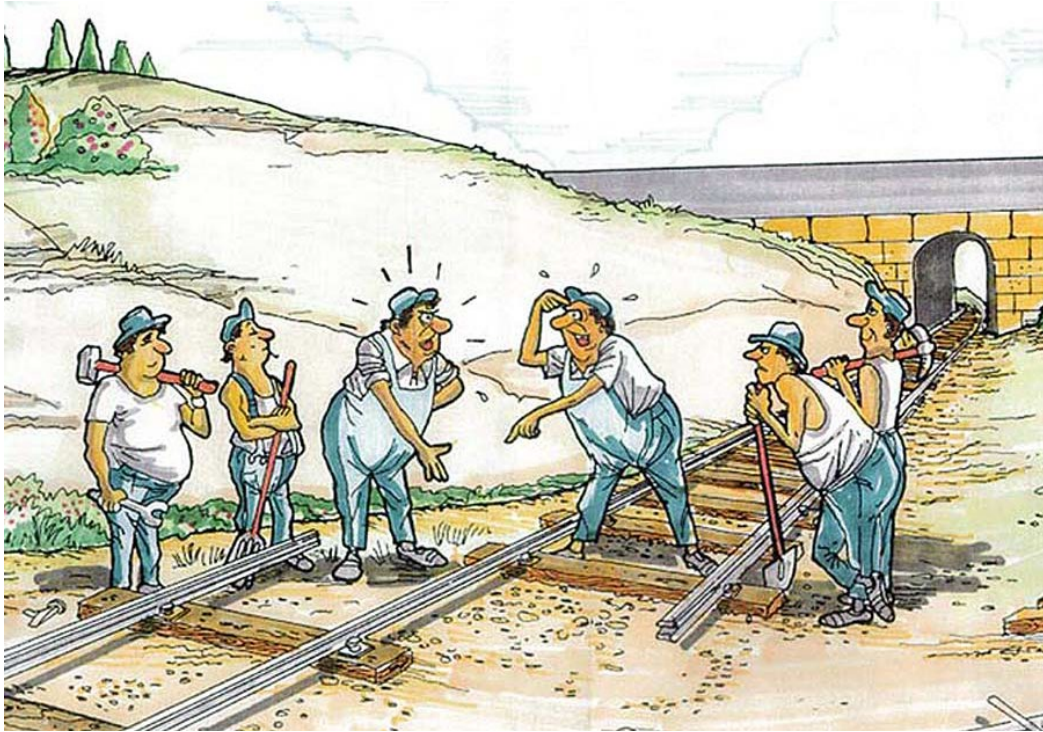
Traditional vs Agile

A few words about management

	Traditional	Agile
Fundamental Assumptions	Systems are fully specifiable, predictable, and can be built through meticulous and extensive planning.	High-quality, adaptive software can be developed by small teams using the principles of continuous design improvement and testing based on rapid feedback and change.
Control	Process centric	People centric
Management Style	Command-and-control	Leadership-and-collaboration
Knowledge Management	Explicit	Tacit
Role Assignment	Individual—favors specialization	Self-organizing teams—encourages role interchangeability
Communication	Formal	Informal
Customer's Role	Important	Critical

Large Projects in Agile

How to upscale smoothly in large projects?

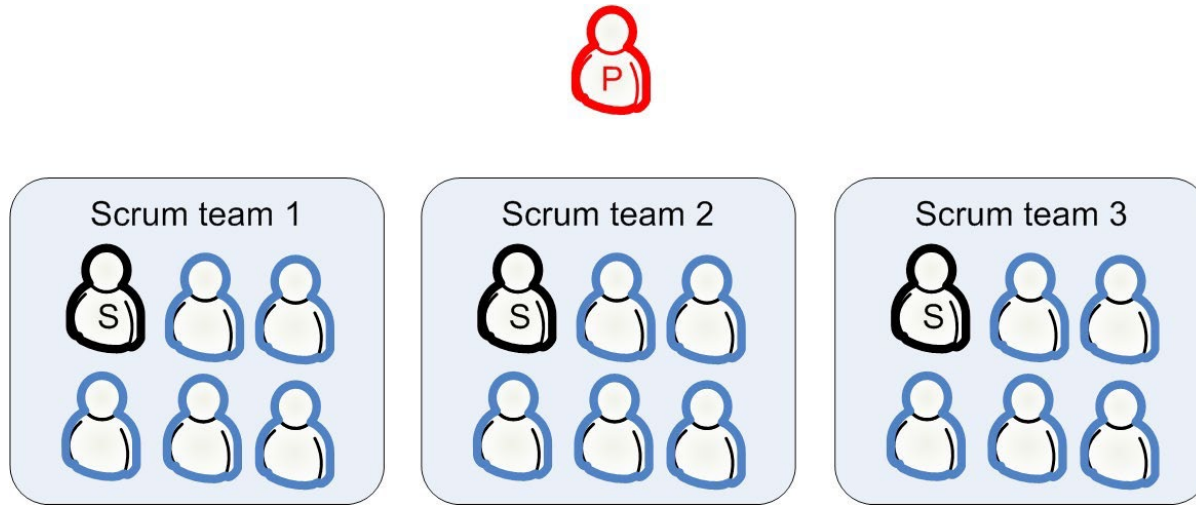


What is large-scale anyway?

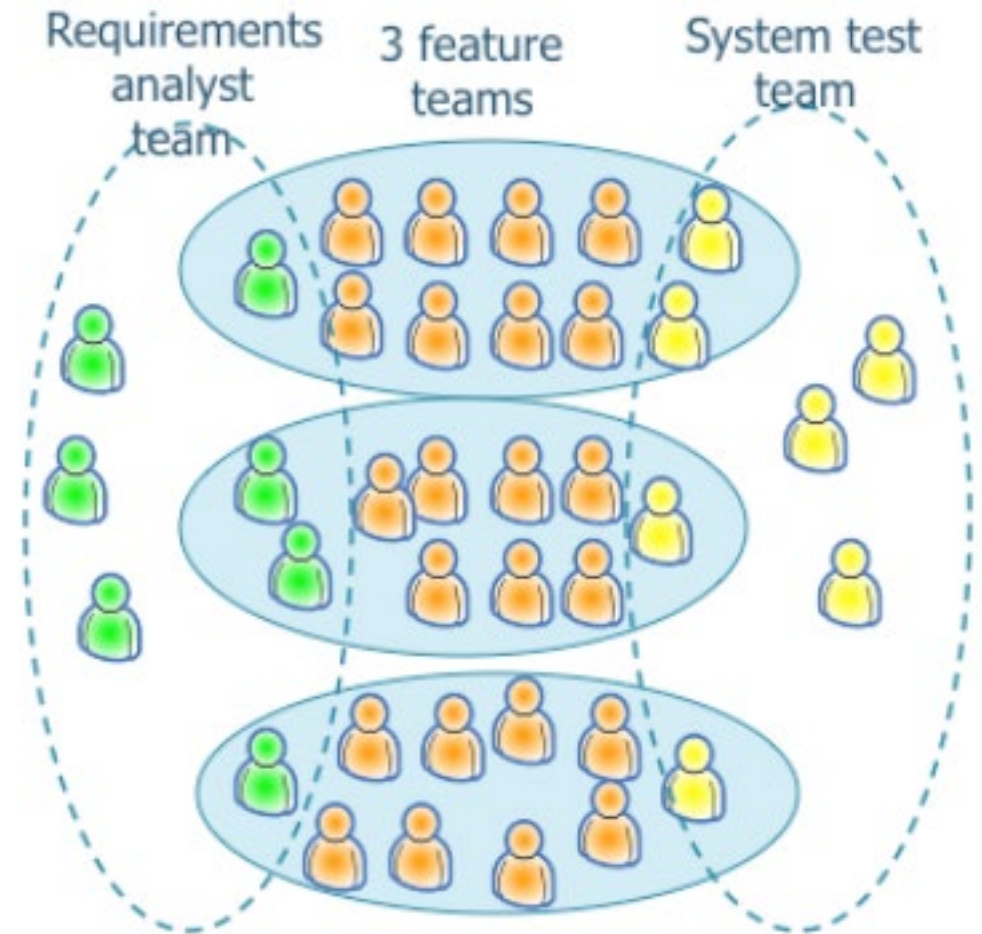
- More than 50-100 developers (Elshamy and Elssamadisy 2006) or more than 20 (Hannay and Benestad 2010)
- More than 100 KLOC (Petersen & Wohlin, 2009)
- The duration of the project (Bjarnasson et al 2011)
- **2-9 development teams (Dingsøy et al 2014)**

Agile in larger projects

How many teams is too many teams?



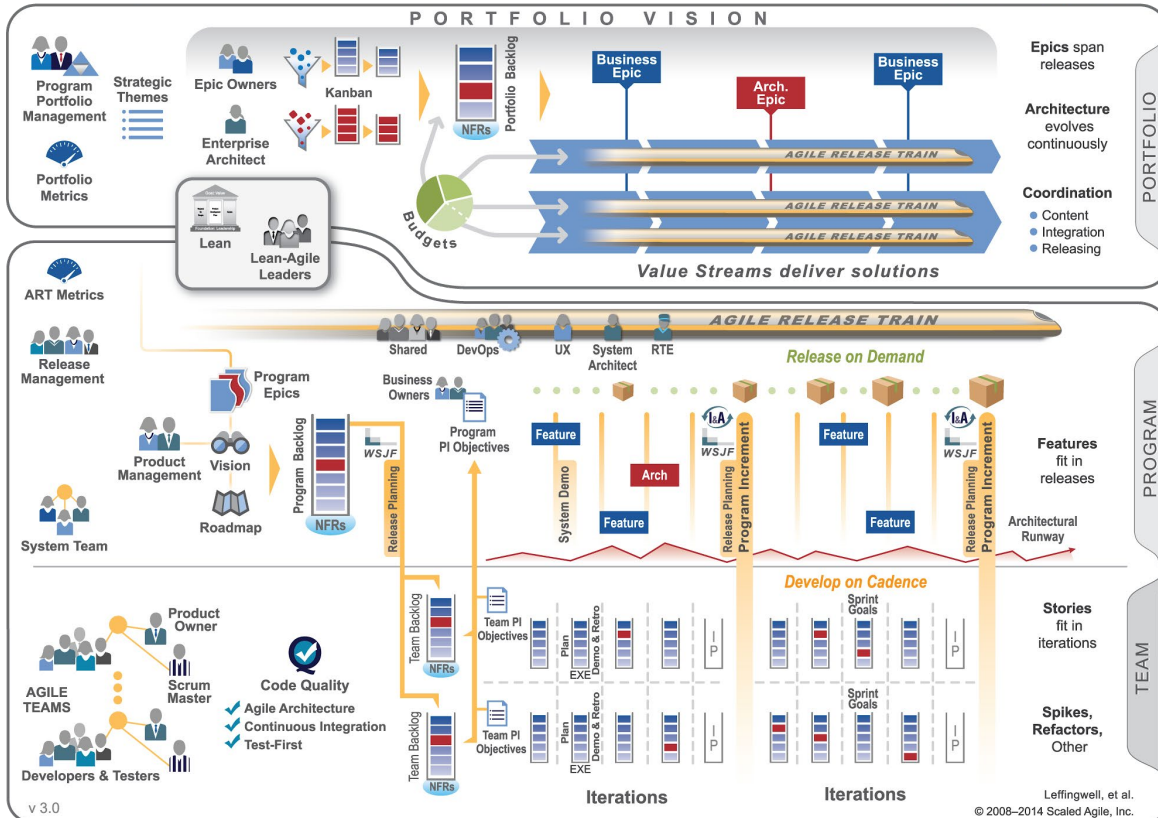
Kniberg, Henrik. **Scrum and XP from the Trenches**. How we do Scrum. InfoQ, (2015).



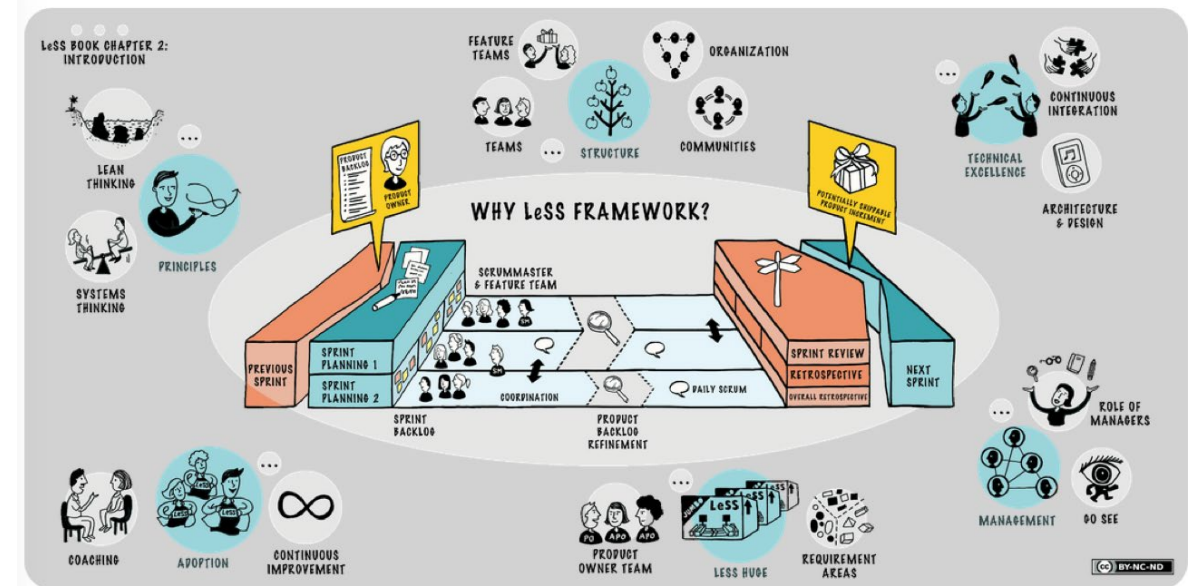
Agile in larger projects

SAFe and LESS

Scaled Agile Framework®



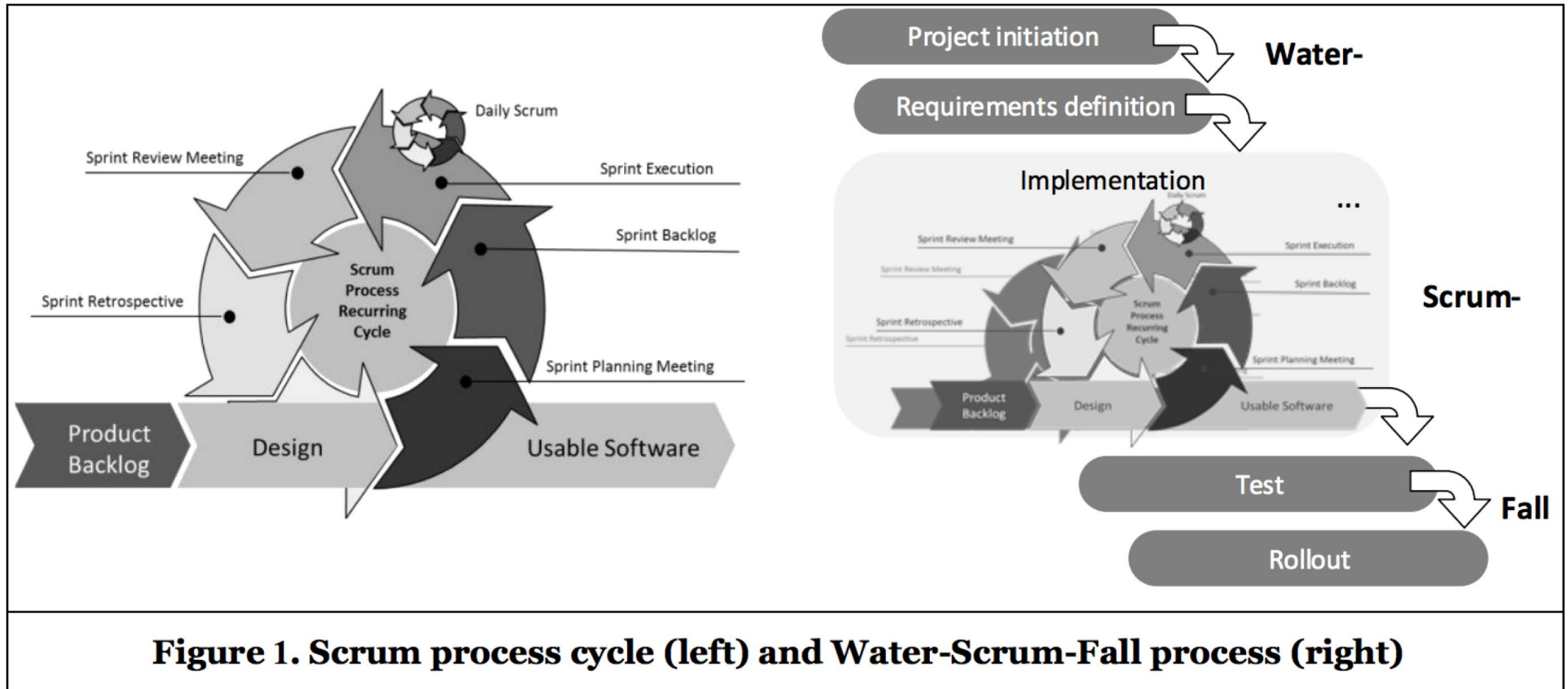
<http://www.scaledagileframework.com/>



www.less.works

Mixed approaches

e.g. Water-Scrum-Fall



Schlauderer, Sebastian, Sven Overhage, and Björn Fehrenbach. **Widely used but also highly valued? Acceptance factors and their perceptions in water-scrum-fall projects.** Proceedings of the 36th International Conference on Information Systems (ICIS), Fort Worth, TX, USA 2015. AIS Electronic Library (AISeL), 2015.

Questions?

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Exercise

Product Backlog

- Create a product backlog for the **first release** of the IS project you are working on
- Use different degrees of detail based on the Agile/Scrum practices
- Current Sprint (Sprint 1):
 - Write user stories with acceptance criteria
 - Prioritize items based on importance/value
- Upcoming Sprints (Sprint 2 and beyond):
 - Add planned PBIs with high-level descriptions
 - Keep these less detailed compared to the current release
 - Themes and epics are allowed beyond the current next release
- **NOTE!** Product backlog reflect the project's Scope in the Iron Triangle.



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